QSIP HACKATHON · EXTREMEDATA4QML · STOCKHOLM · 6 NOV 2026

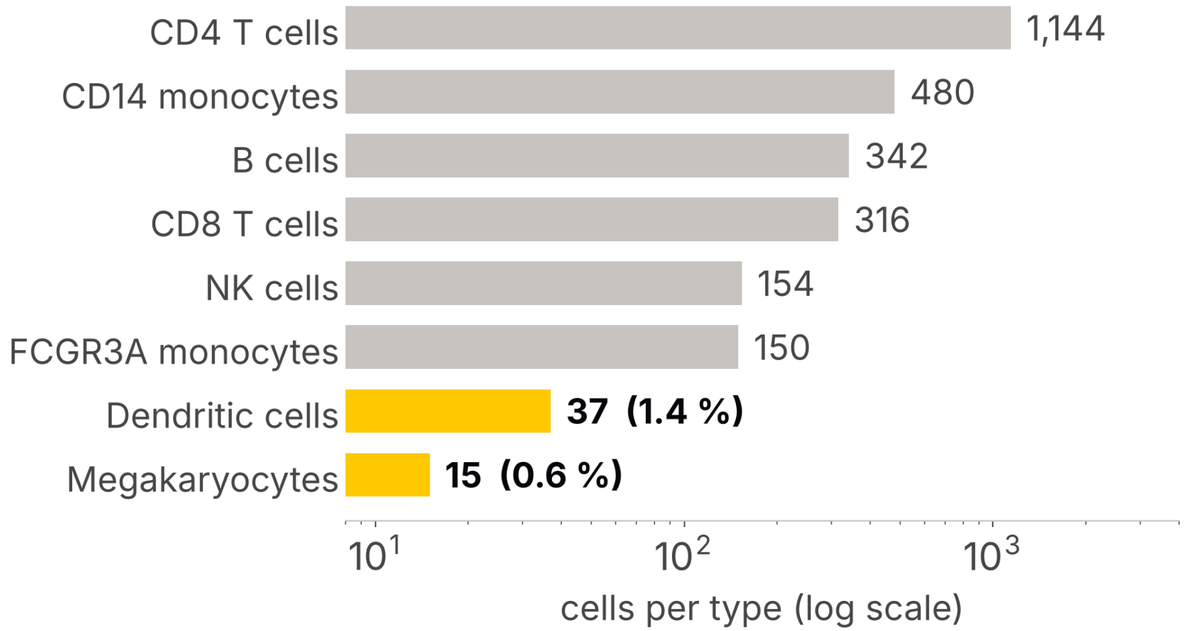
Learning rare cells from a data stream too big to store

Quantum oracle sketching: every record becomes one quantum gate, and is then forgotten.

1 Dataset → 2 Challenge → 3 Quantum idea → 4 Prototype → 5 Real world

STEP 1 · DATASET · STEP 2 · CHALLENGE

Blood cells: mostly zeros, and the key ones are rare



PBMC3k: 2,638 blood cells, 10x Genomics

32,738
genes measured per cell

94 %
of the values are zero

1 in 71
is a dendritic cell; 1 in 176 a megakaryocyte

STEP 3 · THE QUANTUM IDEA

Each record nudges a phase, then it is gone

01

Stream of records

(cell, gene, value, label): read once, never stored, zeros skipped

02

One phase gate each

All gates commute, so the register keeps only their sum

03

One measurement

A Hadamard test reads out a score: rare or common

$$n = \lceil \log_2 N \rceil \qquad \varphi_g = \frac{\tau}{M} \sum_{t=1}^{M} w_t \, [\, g_t = g \,] \qquad \langle Z \rangle = \sum_g a_g^2 \sin \varphi_g$$

Like rain on drums: no drum keeps a single drop, but each ends up tuned by how much fell on it.

STEP 3 · WHY IT FITS THIS DATA

Memory stays tiny; rare classes cost samples, not qubits

Memory: few qubits; price: more records

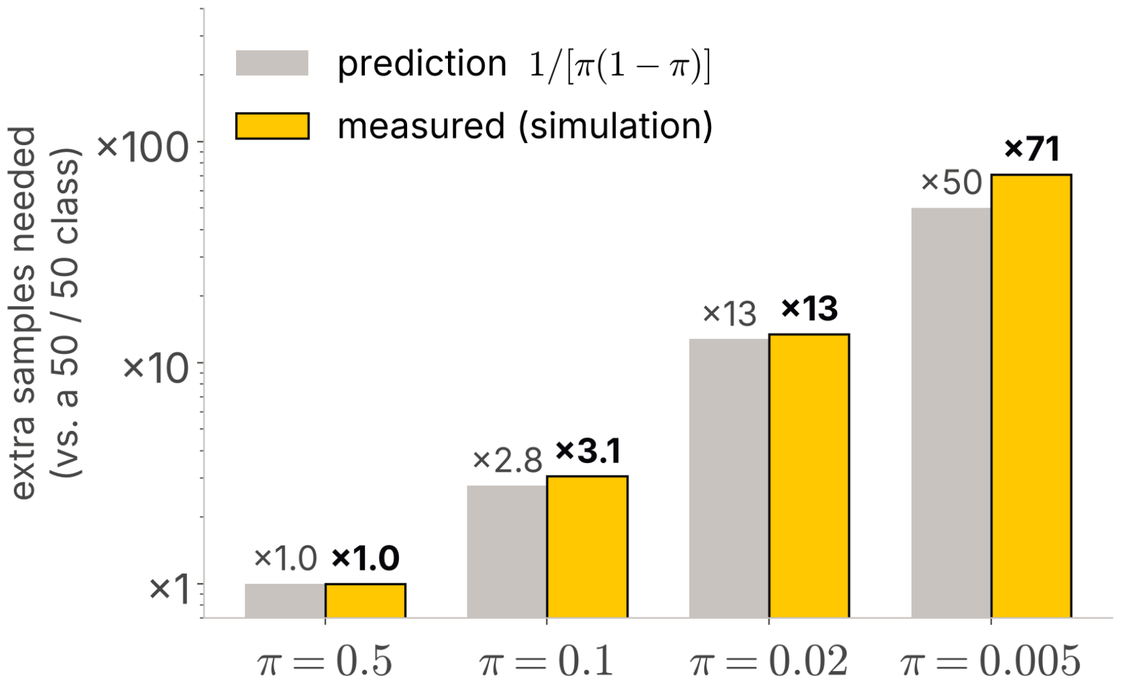
$$\varepsilon^2 \approx C\,\frac{N}{M}$$

Error falls as records stream in; measured with no fitted parameter.

Our addition: fair to rare classes

$$M \propto \frac{1}{\pi\,(1-\pi)}$$

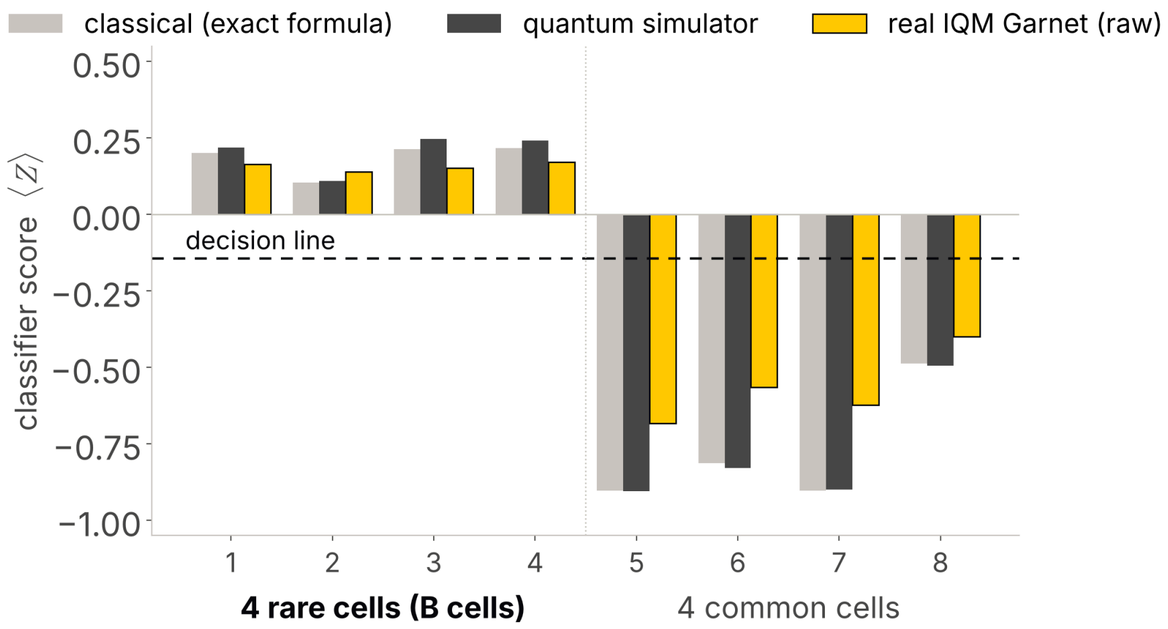
Weighted gates keep rare classes fair; they pay in records: dendritic cells need ×72.



Synthetic test: same data, only the rare share changes

STEP 4 · PROTOTYPE

It runs on real quantum hardware: 8 of 8 test cells right



Same 8 held-out cells, same frozen model · 2,000 shots

device	3 qubits	5 qubits
IQM Garnet	8 / 8 right	5 / 8 → 8 / 8
signal kept	78 %	43 %
IBM Kingston	8 / 8 right	8 / 8 right
signal kept	89 %	74 %

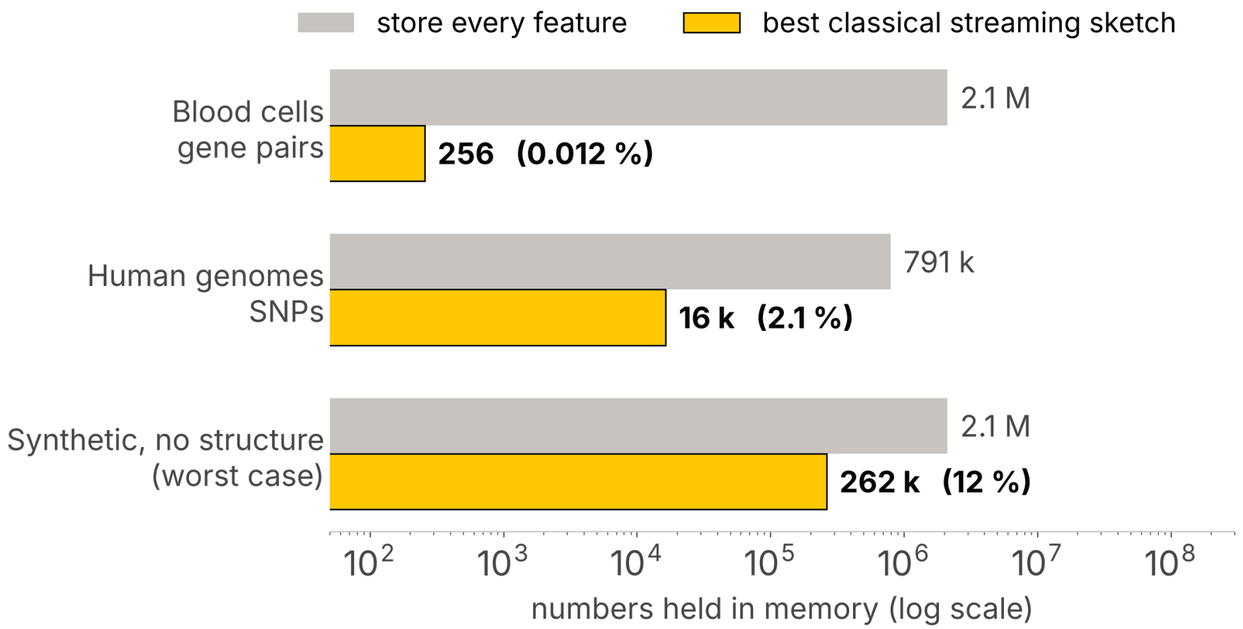
Noise shrinks the score. Five known circuits measure the shrink and undo it:

$$z_{\text{hw}} = \alpha\, z_{\text{ideal}} + \beta$$

Trained in simulation; the hardware runs inference only.

HONEST CHECK · COULD CLASSICAL DO IT WITH LITTLE MEMORY?

Yes: on real data, a small classical sketch is enough



Memory needed to match the full classifier · 5 seeds per point

Blood cells

The signal sits in a few marker genes: 256 numbers do the job.

Human genomes

Signal spread over many SNPs, but compressible: memory stays flat as data grows.

Only structureless data

forces classical memory to grow with the data. That is where the quantum bound lives.

STEP 5 · REAL WORLD

Where this matters, and what it would really cost

Cell atlases

100 million cells, arriving as a stream

Liquid biopsies

A tumour cell is 1 in a million blood cells

Fraud and money laundering

0.1–0.2 % of transactions, endless streams

one rare-cell sketch	all genes	all gene pairs
quantum memory	30 logical qubits	60 logical qubits
quantum time	≈ 11 minutes	≈ 1.5 years
classical memory	131 kB	4.3 GB

Fault-tolerant gates at 1 MHz, 10 % error, dendritic-cell rarity

What we show

A working mechanism with measured costs, running on two real QPUs. Not a quantum advantage: that needs data that is dense **and** incompressible.